Operating Manual

Version 1.11.1

SENSO-Stick

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Revisions

Date	*Version*	*Description*
17.05.2010	1.10	Customization on DLR SENSO-Stick with two CAN bus systems.
23.03.2015	1.11.0	Operating manual of the revised SENSO-Stick. Division into operating manual and software manual.
18.03.2016	1.11.1	Added English translation. Updated contact data. Several smaller changes.

1 Introduction

1.1 About this Operating Manual

- Prerequisite for safe handling and failure-free use of the SENSO-Stick is the knowledge of the basic safety rules and regulations.
- This Operating Manual contains the most important information on how to use the SENSO-Stick safely.
- All personnel using the SENSO-Stick are to observe this Operating Manual as well as the Software Manual.
- Moreover the rules and regulations in effect at the specific place of operation are to be observed.
- This manual is currently available in:
  - German (original manual)
  - English (translation of the original manual)

1.2 Designated Use

The SENSO-Stick Stick is a 3-axis input device with integrated drives for generating the reacting forces (force feedback joystick). It is exclusively designed for using in simulation, training, research and development. Any other use or any use beyond that is misuse. Using the device in steering systems for vehicles employed in regular road traffic is strictly forbidden. SENSODRIVE GmbH cannot be held liable for any damages resulting of such usage.

Designated use includes paying regard to all information in this operating manual.

1.3 Product variants and types

This operating manual is designated for the type:

SENSO-Stick – Standard

1.4 Scope of Supply

Tabletop device complies of:

– SENSO-Stick
– CAN cable, terminated with 120Ω at one end
– Power supply cable

Optional:

– CAN-USB interface and driver software
– Power supply cable with line filter
– Power supply unit

2 Technical Data

2.1 Functional Data SENSO-Stick

Range of motion X-/Y-/Z-axis	± 20°/± 20°/± 15°
Maximum force X-/Y-axis	32 N
Nominal force X-/Y-axis	11 N
Maximum Torque Z- axis	2 Nm
Weight of tabletop device	≈ 12 kg

2.2 Functional Data Motor

Nominal torque	0.7 Nm
Maximum torque	2.0 Nm
Torque resolution	M_{Δ} = 0.03 Nm
Maximum speed	$n_{max} \approx 200\ min^{-1}$
Encoder resolution	24000 inc/360°

The corresponding values can also be read out via CAN (see Software Manual, Chapter 7.3, System Parameters).

2.3 Dimensions

The following Fig. 2-1 shows the external dimensions of the SENSO-Stick:

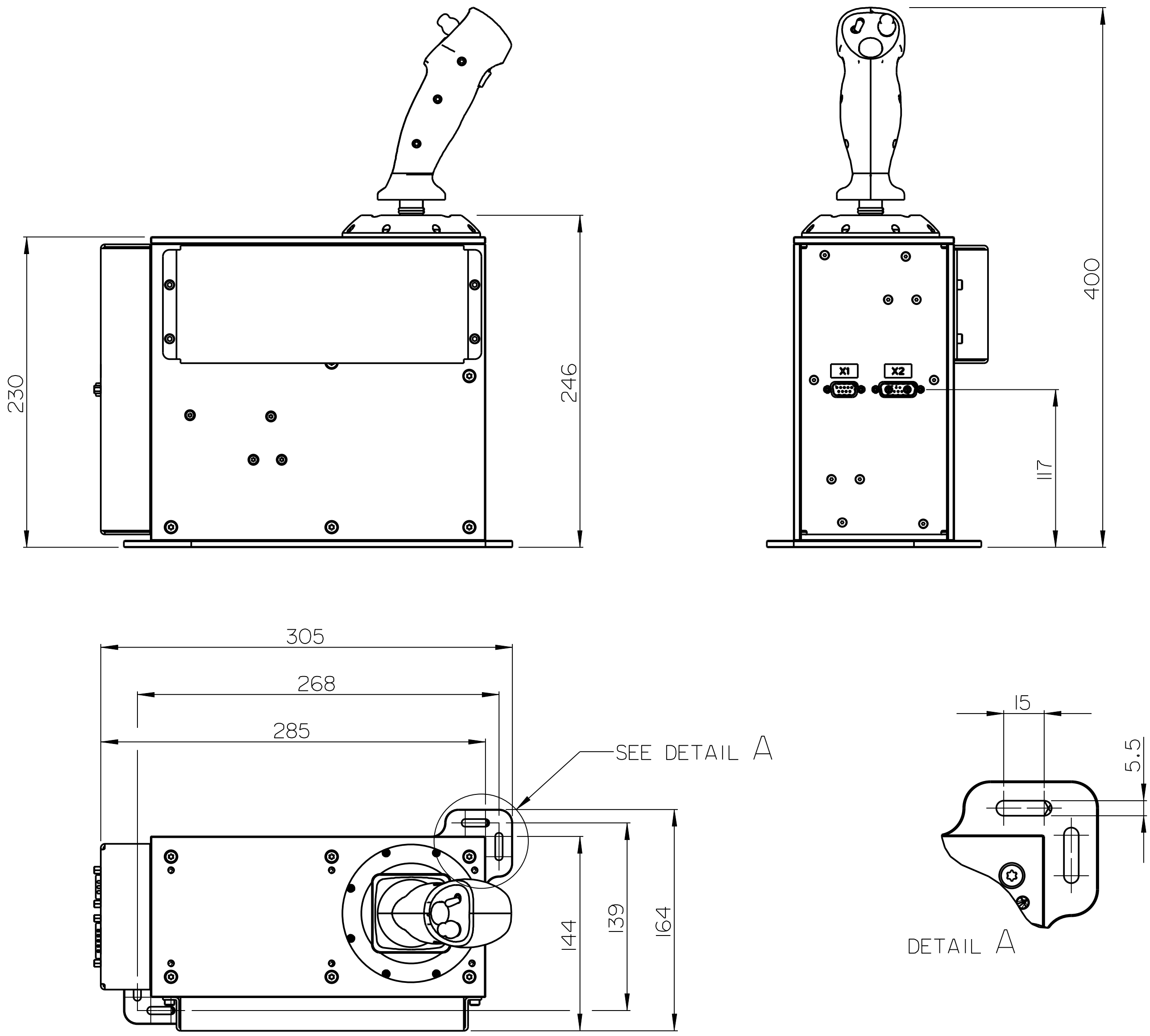

Fig. 2-1: Dimensions of the SENSO-Stick, all dimensions in mm

2.4 Pin Assignment

On the back of the SENSO-Stick there are two plug-in connections to the power supply and communication via CAN. Fig. 2-2 shows the arrangement of the contacts.

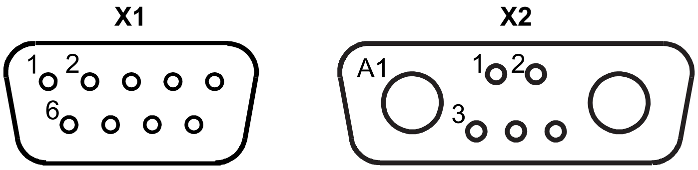

Fig. 2-2: Conntector on the back

The assignment of the two connectors can be found in the table below:

[X1]: CAN	
Function	**Terminal [X1]**
Shield	Housing SENSO-Stick
-	X1 - 1
CAN low	X1 - 2
CAN GND	X1 - 3
-	X1 - 4
-	X1 - 5
-	X1 - 5
CAN high	X1 - 7
-	X1 - 8
-	X1 - 9

[X2]: Power Supply	
Function	**Terminal [X2]**
Shield	Housing Controller
+48V Output stage	X2 - A1
0V (GND)	X2 - A2
+48V Logic	X2 - 1
0V (GND)	X2 - 2
-	X2 - 3
-	X2 - 4
-	X2 - 5

2.5 Buttons and Switches

The handle of the SENSO-Stick has two buttons, a toggle switch and a mini stick. Fig. 2-3 shows the positions of the switches and buttons on the joystick handle of the SENSO-Stick. Also see Chapter 6.6 in the Software Manual – Reading the buttons and switches on the joystick handle.

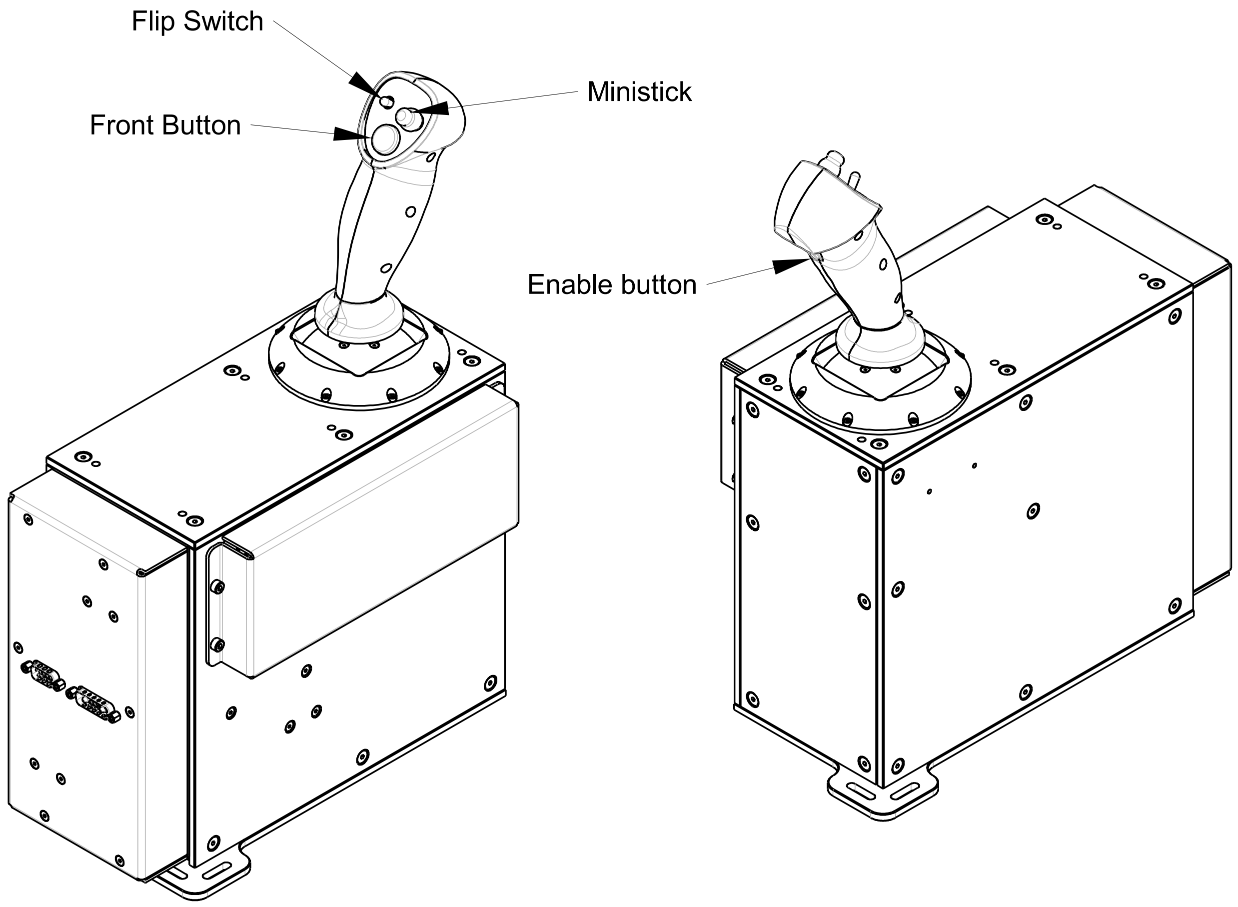

Fig. 2-3: Buttons and switches on the SENSO-Stick

2.6 Communication Interface

Communication with the SENSO-Stick by

CAN.

Preset transfer rate of

1 Mbit/s = 1 Mbaud.

2.7 Definition of the Axes

Fig. 2-4 shows the top view of the SENSO-Stick. Located in this are the movement directions of the handle.

An upward movement of the handle corresponds to a positive movement in the y-direction. If you move the handle to the right, this represents a positive movement in the x-direction. Upon rotation of the handle counterclockwise to obtain the positive angle of rotation z.

A positive motor torque rotates the SENSO-Stick in a positive direction.

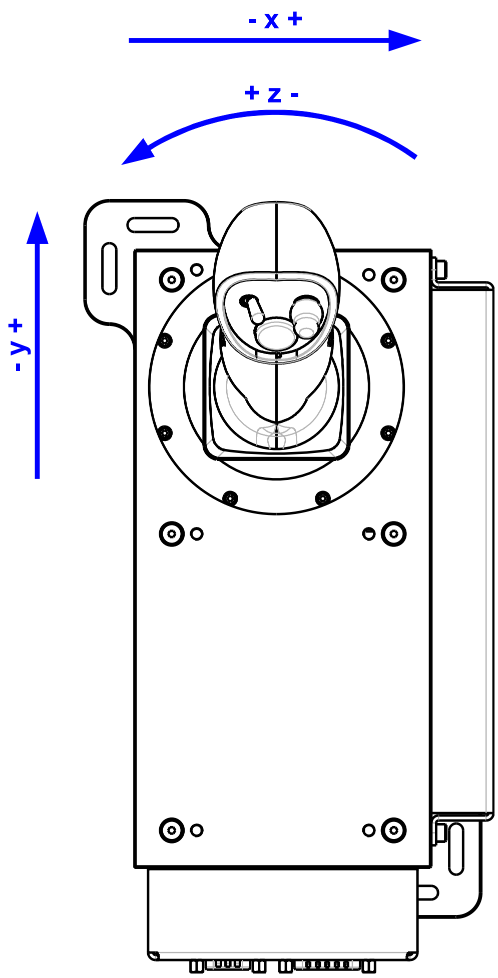

Fig. 2-4: Movement directions of the SENSO-Stick

2.8 Error of the Handle Position and the Torque of the z-Axis

Due to the design of the z-axis when the handle is deflected a difference arises between the motor position and the rotation angle of the handle, as well as the motor torque and the torque of the handle.

The angle difference between input and output $z_{handle} - z_{motor}$ depends on the deflection of the handle with a maximum of ±1.8°. The maximum torque deviation between $\tau_{z,\,motor}$ and $\tau_{z,\,handle}$ amounts to ±7%.

2.9 Environmental Conditions

Operating temperature	+10°C to +30°C
Storage temperature	+0°C to +50°C
Humidity	max. 70% non-condensing

	Only to be used in dry, closed rooms.

2.10 Electrical Connection Values

Power connection at SENSO-Stick:

Supply voltage:	24 … $48V_{DC}$
Typical maximum supply current at $24V_{DC}$ / $48V_{DC}$:	15A / 7.5A
Absolute maximum supply current at $24V_{DC}$ / $48V_{DC}$:	24A / 12A

	Only use a certified power supply unit with secure isolation according to EN 61010-1 or EN 60950.

3 Safety Instructions

3.1 Designated Use

Please refer to chapter 1.2 for information on the designated use of the SENO-Stick.

3.2 Instructions on Operation and Use

Operator, in scope of this document, is anyone who integrates the SENSO-Stick in a simulator and/or operates in conjunction with a host controller (host). In contrast, the user is the one who has his hand on the joystick handle of the SENSO-Stick. The operator is responsible for the proper use of the SENSO-Stick according to the directions stated in Chapter 3.2.2.

3.2.1 Instructions on the Operation of the SENSO-Stick

- The SENSO-Stick is not an apparatus in the sense of the “Law on Equipment Safety”, the “Law on EMC” or the “EG Machinery Directive”. It is a component. It is only to be used in conjunction with a host. The SENSO-Stick may not be put into operation until the whole apparatus that the SENSO-Stick has to be built into, has been put together and is in accordance with the European Directives.
- It is the operator's responsibility to ensure the compliance with the national safety regulations of the combination of the SENSO-Stick together with the host.
- Do not use the SENSO-Stick without monitoring the communication between the host and the SENSO-Stick. The host should monitor the communication at all times (Host Watchdog).
- On no accounts should the safety functions of the SENSO-Stick be de-activated.
- It is the responsibility of the operator to make sure that the SENSO-Stick behaves like expexted.
- Limit the torque to 20% while the SENSO-Stick is put into operation (see Software Manual).

3.2.2 Instructions on the Use of the SENSO-Stick

- Inexperienced users must be supervised while using the SENSO-Stick.
- It is forbidden to put fingers, hands or any other body parts into the stop area of the joystick handle. The stopper area is located below the handle between the plastic stopper and the metal shaft of the handle. It limits the range of movement of the joystick handle.
- Children below the age of fourteen are not allowed to use the SENSO-Stick.

3.3 Device Specific Instructions

– In case of emergency the SENSO-Stick has to be disconnected quickly and reliably from the supply. An emergency switch shall be provided.

	**WARNING** The contact rating of the emergency switch must match the supply voltage and current.

Voltage	Current
$230V_{AC}$ / 50Hz	3A
$24V_{DC}$ / $48V_{DC}$	24A / 12A

– Mount the emergency switch no further than one meter from the SENSO-Stick at a fixed location.

– Use only certified power supplies with isolation according to EN 61010-1 or EN 60950.

Non-observance of following instructions can cause serious injury.

	**WARNING** Automatic moving parts The SENSO-Stick handle moves automatically when instructed by the host controller.

	**WARNING** There is a potential risk to fingers and hands (particularly in the end stop area). Keep hands and arms away

	The mains plug or the main switch must be within easy reach to allow the SENSO-Stick to be disconnected quickly from the power supply.

	The SENSO-Stick is to be placed on a stable structure and attached accordingly, e.g. table top.

4 System description of the SENSO-Stick

The SENSO-Stick consists of three brushless DC motors (SENSO-Stick Motor) each having a control device (SENSO-Stick Controller). Fig. 4-1 shows schematically the construction of modules and the integrated version of the SENSO-Stick.

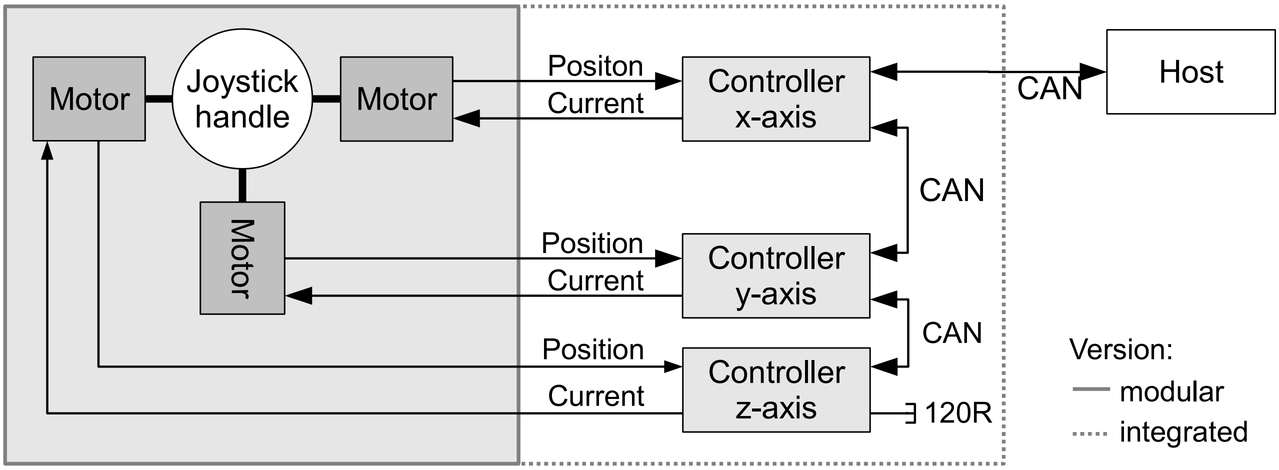

Fig. 4-1: Movement directions of the SENSO-Stick

5 Initial Commissioning

In this chapter it is explained how to establish the necessary cable connections.

Please refer to the Software Manual for an initial CAN communication with the SENSO-Stick.

5.1 CAN and Power Cable

5.1.1 CAN Cable

The wiring of the CAN cable is shown in Fig. 5-1.

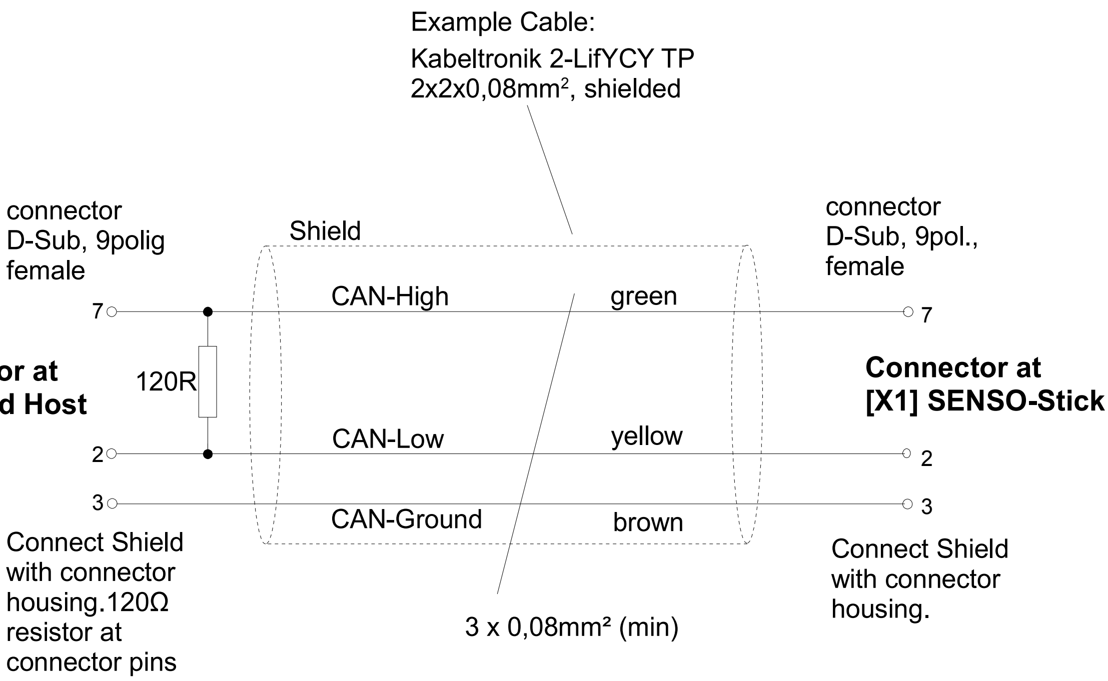

Fig. 5-1: CAN cable

	NOTICE To avoid reflections, the CAN bus must be terminated with 120Ω resistor at the connector of the host CAN card. The opposite side is internally terminated with a 120Ω resistor.

5.1.2 Power Cable

The wiring of the power cable is shown in Fig. 5-2.

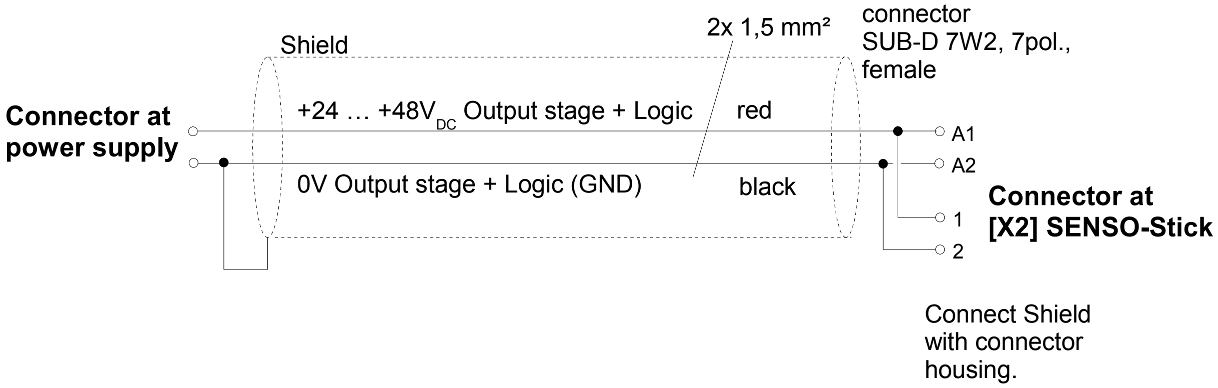

Fig. 5-2: Power cable for joint supply of logic and power

If, for safety reasons, you want to separate the power and logic, then Fig. 5-3 shows how the wiring is carried out.

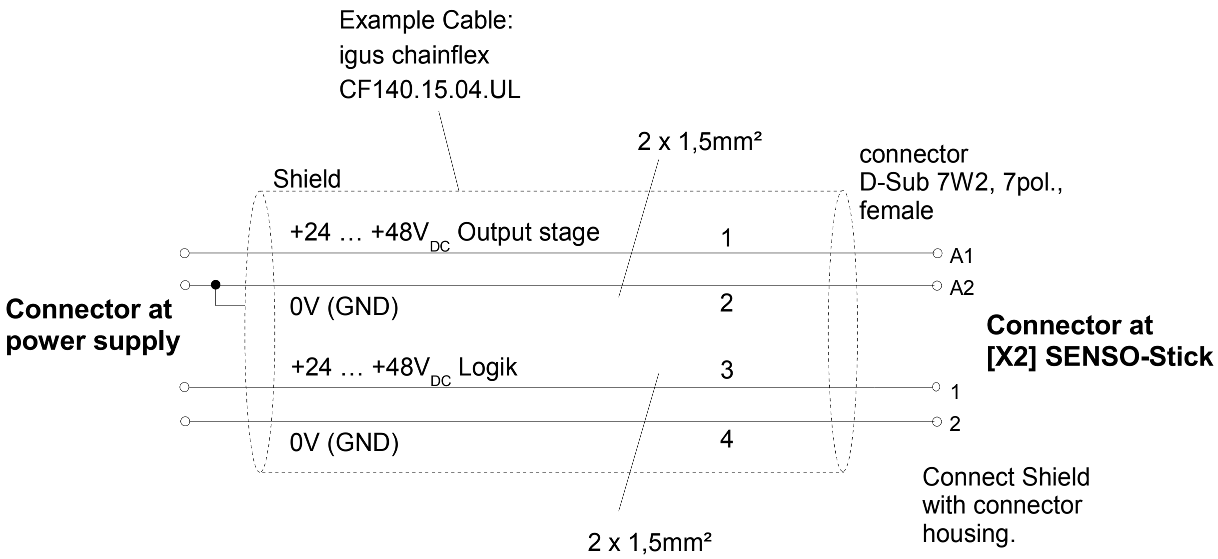

Fig. 5-3: Power cable for separate the supply of power and logic

5.2 Manufacture of Connectors and Power Supply

1) Connect CAN cable with connector [X1] of the SENSO-Stick.
2) Connect CAN cable with the host.
3) Connect supply cable with connector [X2] of the SENSO-Stick.
4) Connect the power supply cable to a non-energized source of 24 … $48V_{DC}$.
5) Turn supply voltage on.

Communication with the SENSO-Stick and the consequent functions are described in detail in the SENSO-Stick Software Manual.

6 Cleaning

	Cleaning is not necessary! If required, wipe the SENSO-Stick with a dry cloth only. Do not use any solvents or chemicals.

7 Repair

In case of defect please contact:

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8 Disposal

	NOTICE The SENSO-Stick materials shall be separated and disposed of in compliance with the local waste disposal act. Follow the national regulations!